Week 6 Dosage Calculation Activity

Oral & IV Medications · Conversions · Intake & Output

Name:	Date:	Section:
_____	_____	_____

PATIENT OVERVIEW: HENRIETTA LOU

You are caring for Henrietta Lou, a 75-year-old female admitted for treatment of a worsening venous stasis ulcer on her right lower extremity. She also has a history of COPD, chronic venous insufficiency, DVT, obesity, and anxiety. Her weight is 69.5 kg. She has NKDA (no known drug allergies). Use the information below to answer all questions in this activity.

Weight	69.5 kg	**Age**	75 years
Height	5' 5"	**Allergies**	NKDA
Diagnosis	Venous stasis ulcer (R lower extremity); COPD; DVT history	**Code Status**	Full Code

Active Medication Orders

Medication	Order	Available (on hand)	Route
Acetaminophen	325–650 mg	325 mg tablets	PO, q6h PRN pain
Aspirin	81 mg	81 mg tablets	PO, Daily
Zinc supplement	1 tablet	220 mg tablets (elemental zinc 50 mg)	PO, Daily
Multivitamin	2 tablets	Standard multivitamin tablets	PO, Daily
Albuterol/Ipratropium	100 mcg–20 mcg per puff	MDI inhaler	Inhaled, PRN wheeze, max 6 doses/24hr
NS IV Fluid	1,000 mL	0.9% Normal Saline bag	IV via pump, over 8 hrs
Vancomycin	1,000 mg	500 mg/100 mL piggyback	IVPB via pump, over 90 min
Furosemide (Lasix)	40 mg	10 mg/mL vial (IV)	IV push

Morphine sulfate	4 mg	10 mg/mL vial	IV push PRN pain

SECTION 1: CONVERSIONS

Question 1. Henrietta weighs 69.5 kg. Convert her weight to pounds. Round to the nearest tenth.

Answer: 152.9 lb | Work: 69.5 kg × 2.2 lb/kg = 152.9 lb

Question 2. The provider receives a call and wants to know Henrietta's weight in grams. Convert 69.5 kg to grams.

Answer: 69,500 g | Work: 69.5 kg × 1,000 g/kg = 69,500 g

Question 3. Albuterol is ordered at 100 mcg per puff. Convert 100 mcg to milligrams.

Answer: 0.1 mg | Work: 100 mcg ÷ 1,000 mcg/mg = 0.1 mg

Question 4. Henrietta drank 6-oz of chicken broth with her lunch. Convert 6 oz to milliliters.

Answer: 180 mL | Work: 6 fl oz × 30 mL/fl oz = 180 mL

Question 5. For her dinner tray, the dietary aide records that Henrietta had 1.5 cups of soup. Convert 1.5 cups to milliliters for the I&O record.

Answer: 360 mL | Work: 1.5 cups × 240 mL/cup = 360 mL

SECTION 2: ORAL MEDICATIONS

Question 6. Henrietta rates her pain as a 5/10. You decide to give her the maximum ordered dose of acetaminophen. Available: 325 mg tablets. How many tablets will you administer?

Ordered: 325–650 mg | Available: 325 mg tablets | Maximum single dose = 650 mg

Answer: 2 tablets | Work: 650 mg ÷ 325 mg/tablet = 2 tablets

Question 7. The provider adds a new oral antibiotic order: Amoxicillin 500 mg PO q8h. The pharmacy sends 250 mg capsules. How many capsules do you give per dose?

Ordered: 500 mg | Available: 250 mg capsules

Answer: 2 capsules | Work: 500 mg ÷ 250 mg/capsule = 2 capsules

Question 8. A liquid medication is ordered: "0.25 g PO." Available: 100 mg/5 mL oral suspension. How many mL will you administer? Round to the nearest whole number.

Answer: 13 mL | Work: 0.25 g × 1,000 = 250 mg; (250 mg ÷ 100 mg) × 5 mL = 12.5 mL; round to whole mL = 13 mL

SECTION 3: IV MEDICATIONS & PUMP RATES

Question 9. Henrietta has an order for 1,000 mL of 0.9% NS to infuse over 8 hours via IV pump. What rate will you set on the pump?

Answer: 125 mL/hr | Work: 1,000 mL ÷ 8 hr = 125 mL/hr

Question 10. Vancomycin 1,000 mg (in 100 mL NS piggyback) is ordered to infuse over 90 minutes via pump. What pump rate in mL/hr will you set? Round to the nearest whole number.

Answer: 67 mL/hr | Work: 90 min ÷ 60 = 1.5 hr; 100 mL ÷ 1.5 hr = 66.7 → 67 mL/hr

Question 11. The furosemide (Lasix) 40 mg IV push is ordered. Available: 10 mg/mL vial. How many mL will you draw up?

Answer: 4 mL | Work: 40 mg ÷ 10 mg/mL = 4 mL

Question 12. Morphine sulfate 4 mg IV push is ordered for pain. Available: 10 mg/mL vial. How many mL will you draw up? Round to the nearest tenth.

Answer: 0.4 mL | Work: 4 mg ÷ 10 mg/mL = 0.4 mL

Question 13. At 0800 you hang a new 1,000 mL NS bag running at 125 mL/hr. What time will the bag finish? (Express as military time.)

Answer: 1600 | Work: 1,000 mL ÷ 125 mL/hr = 8 hr; 0800 + 8 hr = 1600

SECTION 4: GRAVITY DRIP RATES

Question 14. The IV pump is unavailable. You must administer Henrietta's 1,000 mL NS over 8 hours via gravity using 15 gtts/mL tubing. How many drops per minute will you count?

Answer: 31 gtt/min | Work: (1,000 mL × 15 gtt/mL) ÷ (8 × 60 min) = 31.25 → 31 gtt/min

Question 15. You are administering 250 mL of an antibiotic IVPB via gravity over 30 minutes using 20 gtts/mL tubing. How many drops per minute?

Answer: 167 gtt/min | Work: (250 mL × 20 gtt/mL) ÷ 30 min = 166.67 → 167 gtt/min

SECTION 5: INTAKE & OUTPUT (3 PTS EACH)

Question 16. Calculate Henrietta's total INTAKE for the following items from the 0700–1500 shift. Show each conversion and your final total.

Item	Amount Given	mL
1,000 mL NS via IV pump	1,000 mL	1,000 mL
Orange juice with breakfast	6 oz	180 mL
Cup of broth (lunch)	1 cup	240 mL
Cup of ice chips (dinner)	240 mL cup	120 mL (half of 240 mL ice chips)
Liquid acetaminophen (oral)	15 mL	15 mL
Water (sips throughout shift)	4 oz	120 mL
TOTAL INTAKE	1,675 mL	

Question 17. Calculate Henrietta's total OUTPUT for the same 0700–1500 shift.

Item	Amount	mL
Foley catheter urine	550 mL	550 mL
Wound drain (right leg)	2 oz	60 mL
Emesis (nausea after meds)	90 mL	90 mL
TOTAL OUTPUT	700 mL	

Answer key:

#	Question (abbreviated)	Answer / Key Steps
1	69.5 kg → lbs	69.5 × 2.2 = 152.9 lbs
2	69.5 kg → grams	69.5 × 1,000 = 69,500 g
3	100 mcg → mg	100 ÷ 1,000 = 0.1 mg
4	6 oz → mL	6 × 30 = 180 mL
5	1.5 cups → mL	1.5 × 240 = 360 mL
6	Acetaminophen max dose	650 ÷ 325 = 2 tablets
7	Amoxicillin 500 mg / 250 mg capsules	500 ÷ 250 = 2 capsules
8	0.25 g / 100 mg per 5 mL	0.25 g = 250 mg · 250÷100×5 = 12.5 → 13 mL
9	1,000 mL NS over 8 hrs pump rate	1,000 ÷ 8 = 125 mL/hr
10	Vancomycin 100 mL over 90 min	90 min = 1.5 hr · 100 ÷ 1.5 = 66.7 → 67 mL/hr
11	Furosemide 40 mg / 10 mg/mL	40 ÷ 10 × 1 = 4 mL
12	Morphine 4 mg / 10 mg/mL	4 ÷ 10 × 1 = 0.4 mL
13	1,000 mL at 125 mL/hr — finish time	1,000 ÷ 125 = 8 hrs · 0800 + 8 = 1600
14	Gravity: 1,000 mL / 8 hrs / 15 gtts/mL	(1,000×15) ÷ 480 = 31.25 → 31 gtts/min
15	Gravity: 250 mL / 30 min / 20 gtts/mL	(250×20) ÷ 30 = 166.7 → 167 gtts/min
16	Total intake 0700–1500	1,000+180+240+120+15+120 = 1,675 mL
17	Total output 0700–1500	550+60+90 = 700 mL

Study Guide: Formulas, Workings & Exam Prep

This supplement follows the conversions, rounding, and answer key provided in the Week 6 activity. Use it to review the method behind each completed answer.

Essential conversion facts

1 kg = 2.2 lb; 1 kg = 1,000 g
1 mg = 1,000 mcg; 1 g = 1,000 mg
1 fluid oz = 30 mL; 1 cup = 240 mL
1 hour = 60 minutes; count ice chips as half their measured volume (as used in this activity)

Four exam formulas

Tablets or capsules = ordered dose ÷ strength per tablet/capsule
Liquid medication (mL) = ordered dose ÷ available dose × available volume (mL)
IV pump (mL/hr) = volume (mL) ÷ time (hours)
Gravity drip (gtt/min) = volume (mL) × drop factor (gtt/mL) ÷ time (minutes)

Intake and output: complete working

Intake: NS 1,000 + orange juice (6 × 30) 180 + broth (1 × 240) 240 + ice chips (240 ÷ 2) 120 + oral liquid medication 15 + water (4 × 30) 120 = 1,675 mL.
Output: Foley urine 550 + wound drain (2 × 30) 60 + emesis 90 = 700 mL.
Net fluid balance = intake − output = 1,675 − 700 = +975 mL (additional study calculation; not explicitly asked in the activity).

Important worksheet assumptions

The provided answer key counts the entire 1,000 mL NS bag during the 0700–1500 shift and uses half the stated ice-chip volume. In clinical documentation, count only IV fluid actually infused during the specified shift and follow local I&O policy. The worksheet does not provide a separate infusion-start time for Question 16.
For Question 8, the exact calculated volume is 12.5 mL; the activity specifically requests rounding to the nearest whole mL, producing 13 mL. In clinical practice, follow the ordered precision and institutional medication-administration policies rather than rounding automatically.

Exam-day self-check

Identify the requested unit before calculating.
Convert grams/milligrams/micrograms or minutes/hours before dividing.
Write units in every step and cancel them to verify the setup.
Round only as instructed, typically at the final step.
Check whether a question asks for dose, volume, pump rate, gravity rate, or clock time.
For intake/output, include oral liquids, IV infusions, liquid medications, and measured drainage as directed.